

Stability Degradation Induced by Angle–Voltage Coupling in Power Systems: Bode-Type Fundamental Performance Limitation Analysis

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Abstract—This paper presents a novel perspective on understanding how the interaction between phase-angle and voltage dynamics affects the stability of inverter-dominated power systems. To explore this, we analyze Bode’s integral of frequency-domain sensitivity functions, which reveals the underlying law that limits the system’s ability to reject disturbances. Our finding indicates that, although the “total” sensitivity integrals remain conservative in angle-voltage coupled systems, the coupling reshapes their distribution between phase-angle and voltage dynamics, potentially worsening system stability. This finding is generalized into fundamental laws named *sensitivity conservation and its breaking* in this paper. In addition to the mechanism, our work also suggests that the stability could be improved by appropriately redistributing sensitivity across different frequency bands, diverse devices, or between phase-angle and voltage dynamics. However, this redistribution must be conducted carefully, as the sensitivity breaking law indicates that improving the performance of one aspect may lead to instability of another. This law cannot be bypassed directly by any controllers with a generic feedback structure, underscoring the fundamental limit of angle-voltage coupling dynamics and fostering a broader understanding of its nature. The proposed mechanism provides a core principle for control design in angle-voltage coupled power systems.

Note to Practitioners—This paper introduces a unified approach for assessing fundamental stability and performance limits in power systems with diverse, inverter-based devices. Unlike traditional methods, the proposed one applies broadly without requiring custom models for each device or controller. For engineers, this means system-level trade-offs between frequency and voltage performance can be quantitatively managed, aiding deeper understanding and control designs in complex grids. The method is best suited for small-signal analysis, but it streamlines the evaluation of increasingly heterogeneous and nonlinear networks found in practice.

Index Terms—power system stability, angle-voltage coupling, inverter-dominated power system, sensitivity function, Bode’s

integral, waterbed effect.

I. INTRODUCTION

A. Motivation and Background

INTEGRATING inverter-based resources has significantly altered the stability characteristics of power systems [1], [2]. Due to the weak supportive capability of inverters and their overlapped timescales of phase-angle and voltage dynamics, existing studies have recognized a more strengthened angle-voltage coupling, leading to unprecedented coupled instability for both grid-following (GFL) devices and grid-forming (GFM) devices [3]. Besides, the desynchronization of phase-angles induced by the voltage dip [4] and angle-voltage coupled oscillation [5] are observed more frequently in practice. However, the mechanism of angle-voltage coupling that influences the system dynamics is still unclear. The urgent need for more in-depth investigations of the coupling dynamics primarily motivates this work.

B. Literature Review

Primary research regarding coupled stability focuses on its time-domain feature, mainly based on state space models. The pioneering works [6], [7] account for the voltage dynamics within the synchronization analysis, but set strict conditions for applicable device models. Then, the angle-voltage interaction is investigated from the perspective of the power flow coupling, thus formulating a compatible framework in [8]. Despite its advantages for control enhancement, the coupling mechanism is not quantified. A passivity-based framework is formulated in [9], [10]. By this theory, the angle-voltage coupling strength can be quantified by the exchanged transient energy across phase-angle and voltage loops. However, the construction of storage functions might be tricky and difficult for heterogeneous devices. To address this issue, the literature [11] focuses on the linearized model and derives the damping conservation mechanism, quantifying the coupling’s impact by its strength to disperse eigenvalues. This theory observes in the state-space that the coupling may shift the weakest damping even closer to the imaginary axis, shrinking the stability margin. However, only electromechanical transients are modeled and rigorously proved. Generally, the time-domain methods rely on precise and usually simplified modeling of device and network dynamics, which restricts their applications in gray-box or black-box systems [12].

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In contrast to advances in time-domain studies, frequency-domain-based analysis of angle-voltage coupling has garnered relatively less attention. The latter, however, has valuable interpretations and plays a significant role in the control designs. In addition, contrary to time-domain methods, it is applicable to black-box devices thanks to the impedance scanning techniques. To the authors' knowledge, frequency-domain analysis of angle-voltage coupling in transmission grids is still preliminary. However, a relevant topic is the power decoupling controls extensively studied in power electronics studies. For GFM devices, the power coupling mechanism is comprehensively investigated with a uniform GFM model [13]. The literature indicates that the power interaction comes from the fact that the inverter's terminal voltage is not a global variable but varies locally according to power dynamics, which then feeds back into the phase-angle loops. Therefore, the cross-loop paths should be compensated to suppress the coupled instability [14]. For GFL devices, the cross-loop terms induced by the network interactions are also recognized as a significant factor for coupled instability due to their dual relation with the GFM counterparts [15]. Controllers designed based on the idea of cross-loop path compensation achieve stability enhancement in practice for both GFM devices [16], [17] and GFL devices [18]. While the above methods are restricted to either pure-GFM or pure-GFL systems, the general admittance model is formulated to analyze and suppress the coupled oscillation in hybrid GFM-GFL systems [19], [20]. It is imperative to acknowledge that these studies primarily emphasize the active-reactive power coupling, not angle-voltage coupling. Both topics are closely related since they are all attributed to the network interactions of angle-reactive-power and voltage-active-power cross paths. However, the different input-output selection leads to their distinct dynamics, which makes it necessary to analyze the latter topic in detail.

Pioneering works have guided the control of angle-voltage coupling, yet its dynamics remain underexplored due to the lack of theoretical tools for uniform analysis. Existing literature often requires tailored analyses for specific network topologies or device models. First, existing frequency-domain methods are mainly developed for high R/X distribution grids, assuming parallel networks or using infinite bus representations, which complicates application to meshed transmission systems [21]. However, the coupling can still lead to instability even in inductive networks [11], [17], [22]. Second, these methods focus on the power coupling of inverter-interfaced devices. Despite its close relation, angle-voltage coupling is a distinct issue needing detailed study [10].

C. Main Contribution

Key idea. Prior studies ascertain coupling effects under specific modeling assumptions. This methodology is essential and effective when considering stability estimation. However, the system-by-system methodology also avoids the investigation of the inherent mechanisms of angle-voltage coupling to influence system dynamics. In this work, we address the above challenge using Bode's sensitivity integral [23]. This renowned

technique establishes fundamental performance limitations of dynamic systems that cannot be directly surpassed by control designs within typical feedback structures [24]. Surprisingly, it has received little attention in the community of power system dynamics. For the first time, the paper employs it to explore fundamental difficulties in tackling angle-voltage coupling in power systems. Since this methodology imposes minimal restrictions on system dynamics [25], [26], it can offer generic results that are adaptable to various device dynamics and grid topologies.

Main Contribution. We first extend the multi-input-multi-output version of Bode's integral from singular value to eigenvalue spectrum. Applying the developed theory, it is revealed that the coupling redistributes the sensitivity integral between phase-angle and voltage dynamics and thus influences the system dynamic characteristics, although its summation remains conservative. In addition, the coupling usually shifts the sensitivity toward decreased stability. Compared to existing works, the proposed mechanism leverages its wide applicability to various network topologies and devices with hybrid synchronization mechanisms (GFM or GFL) and distinct control strategies. Therefore, it provides a uniform and generic approach to understanding the coupling dynamics of inverter-dominated power systems. Despite the sensitivity conservation, our results suggest that stability could be enhanced by appropriately redistributing sensitivity across different frequencies, devices, or between phase-angle and voltage dynamics, shedding new light on inverter control designs.

D. Paper Organization

The rest of this paper is organized as follows: Section II introduces the power system models and formulates the problem. The formal demonstration of sensitivity conservation and breaking laws is provided in Section III. The simulation analysis is performed in Section IV on several standard IEEE benchmark systems. In Section V, we extend the proposed theory from pure GFM-integrated systems to hybrid GFM-GFL ones. Finally, Section VI concludes the paper. The article structure is shown in Fig. 1.

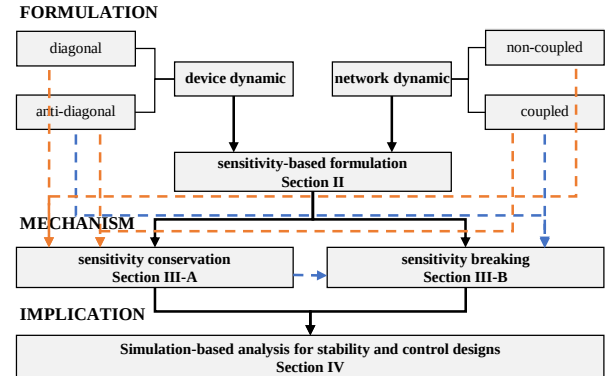


Fig. 1. Structure of the paper.

Notations: For a matrix M , $\lambda(M)$ denotes its eigenvalues and $\sigma(M) = |\lambda(M)|$ for its magnitude. The real and imaginary parts of a complex number c are denoted by $\Re(c)$

and $\Im(c)$, respectively. Throughout the paper, we use the superscript (n) and the subscript n to represent the ideal non-coupled systems.

II. PROBLEM FORMULATION

Consider an n -bus power system $\mathcal{G} = G(\mathcal{V}, \mathcal{E})$, where $\mathcal{V}$ and $\mathcal{E}$ are the set of buses and transmission lines, respectively. Each bus $i \in \mathcal{V}$ has port-wise dynamics $(V_i, \theta_i, P_i, Q_i)$ representing voltage magnitude, phase-angle, active, and reactive power. The vector form $V = [V_1, V_2, \dots, V_n]^T$ is employed and similar for θ, P and Q .

The existence of equilibrium is first assumed throughout the paper. This assumption excludes the instability caused by the vanishment of equilibrium, such as saddle-node or limit-induced bifurcation, which are usually not classified as coupled instabilities [1], [11]. Then, the linearization is conducted as we focus on small-signal stability, which examines dynamic behavior under infinitesimal perturbations around an equilibrium. The large-signal stability analysis is outside the scope of this paper, which relies on Lyapunov techniques; see [27] for examples.

Assumption 1. An isolated equilibrium exists for system $\mathcal{G}$.

A. Network Dynamic Model

The network dynamics are modeled by the power flow equations:

$$\begin{aligned} P_i &= V_i \sum_{j \in \mathcal{V}} V_j (G_{ij} \cos \theta_{ij} + B_{ij} \sin \theta_{ij}) \\ Q_i &= V_i \sum_{j \in \mathcal{V}} V_j (G_{ij} \sin \theta_{ij} - B_{ij} \cos \theta_{ij}) \end{aligned} \quad (1)$$

where $\theta_{ij} = \theta_i - \theta_j$. Linearizing it around an equilibrium yields:

$$\begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} = \begin{bmatrix} \partial P / \partial \theta & \partial P / \partial V \\ \partial Q / \partial \theta & \partial Q / \partial V \end{bmatrix} \begin{bmatrix} \Delta \theta \\ \Delta V \end{bmatrix} \quad (2)$$

where Δ denotes the increment w.r.t. the steady value, i.e., $\Delta P, \Delta Q, \Delta \theta$, and ΔV are deviations in injected active and reactive power, phase-angle, and voltage amplitude, respectively. For brevity, we denote the network coupling matrix as H_{net} with following blocks:

$$H_{\text{net}} \stackrel{\text{def}}{=} \begin{bmatrix} \partial P / \partial \theta & \partial P / \partial V \\ \partial Q / \partial \theta & \partial Q / \partial V \end{bmatrix} \stackrel{\text{def}}{=} \begin{bmatrix} H_{P\theta} & H_{PV} \\ H_{Q\theta} & H_{QV} \end{bmatrix} \quad (3)$$

The matrix H_{net} can be split into two parts $H_{\text{net}}^{(n)}$ and $H_{\text{net}}^{(c)}$, which represent the non-coupled part and angle-voltage coupling, respectively.

$$H_{\text{net}} = \begin{bmatrix} H_{P\theta} & \\ & H_{QV} \end{bmatrix} + \begin{bmatrix} & H_{PV} \\ H_{Q\theta} & \end{bmatrix} \stackrel{\text{def}}{=} H_{\text{net}}^{(n)} + H_{\text{net}}^{(c)} \quad (4)$$

where the quantities with superscripts (n) and (c) represent **non-coupled** and **coupled**, respectively.

B. Device Dynamic Model

The device dynamics vary depending on their control strategies. Due to the proposed method's generality, we use a unified model as shown below:

$$-\begin{bmatrix} \Delta \theta \\ \Delta V \end{bmatrix} = \begin{bmatrix} G_{\theta P}(s) & G_{\theta V}(s) \\ G_{\theta Q}(s) & G_{VQ}(s) \end{bmatrix} \begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} \stackrel{\text{def}}{=} G(s) \begin{bmatrix} \Delta P \\ \Delta Q \end{bmatrix} \quad (5)$$

The matrices $G_{\theta P}(s)$ and $G_{VQ}(s)$ are diagonal. For example, the entry-wise expression of $G_{\theta P}(s)$ is $\text{diag}(G_{\theta P,1}, G_{\theta P,2}, \dots, G_{\theta P,n})$, where $G_{\theta P,i}$ is the phase-angle regulation with its active power injection for the i -th GFM device. In practical inverter controllers, the cross-loop dynamics are usually not considered, i.e., $G_{\theta V}(s) = G_{\theta Q}(s) \approx 0$. So we focus on $G_{\theta P}(s)$ and $G_{VQ}(s)$. Nevertheless, it should be noted that the proposed methodology is sufficiently robust to account for both network-induced and device cross-loop induced couplings.

Both $G_{\theta P}(s)$ and $G_{VQ}(s)$ have two components: one representing the control strategy $\tilde{G}_{\theta P,i}$ and $\tilde{G}_{VQ,i}$ and the other representing the physical lag effect $G_{\text{lag},i}$. Some typical control strategies $\tilde{G}_{\theta P,i}$ and $\tilde{G}_{VQ,i}$ are listed in Table I, the majority of which are first summarized in [28]. After the control strategies selected, the $Q-V$ dynamics of the i -th device can be modeled by $G_{VQ,i} = \tilde{G}_{VQ,i} G_{\text{lag},i}$, where G_{lag} represent the lag effect induced by the inner loops. A similar procedure can also be conducted for $P-\theta$ dynamics to derive $G_{\theta P,i}$.

TABLE I
A BRIEF SUMMARY OF INVERTER CONTROL STRATEGIES [28]

active power control	$\tilde{G}_{\theta P,i}(s)$	reactive power control	$\tilde{G}_{VQ,i}(s)$
$P-\theta$ droop	$\frac{1}{k_p}$	$Q-V$ droop	$\frac{1}{k_q}$
$P-\omega$ droop	$\frac{1}{k_{ps}}$	$Q-\frac{dV}{dt}$ droop	$\frac{1}{k_{qs}}$
improved $P-\omega$ droop	$\frac{1}{k_{ps}} + \frac{1}{k_{pd}}$	improved $Q-V$ droop	$\frac{1}{k_q} + k_{qd}s$
virtual synchronous generator (VSG)	$\frac{1}{(Ms+d)s}$	voltage inertia	$\frac{1}{Js+k_q}$

This paper focuses on the electromechanical models as they are still the primary concern of transmission system operators. The proposed method can be extended to more practical system models, which are summarized below:

- **Consideration of electromagnetic dynamics:** Given the generality of the proposed method, it is easy to incorporate electromagnetic models through a standard transformation from a state-space model to a transfer function model if required. In particular, the inner-loop dynamics, such as current and voltage loops of inverters, can be analogously modeled in $G_{\text{lag},i}$. Examples can be found in [16] and [29]. In the simulations, we have considered the voltage loop dynamics as an example. More details are provided in Appendix B.
- **Extension to hybrid GFM-GFL systems:** The quantities $[\Delta P^T, \Delta Q^T]^T$ and $[\Delta \theta^T, \Delta V^T]^T$ are selected as input and output of transfer functions, respectively. Under such selection, only GFM devices can maintain the properness of $G_{\theta P,i}$ and $G_{VQ,i}$. However, another crucial category of devices, GFL inverters, has an inverse input-output

relationship, i.e., from $[\Delta\theta^T, \Delta V^T]^T$ to $[\Delta P^T, \Delta Q^T]^T$. Simply taking its inverse might induce non-causality of transfer functions and thus does not match the above theoretical framework. This problem serves as a common difficulty for frequency-domain-based analysis methods of angle-voltage coupling when considering hybrid GFM-GFL integrations. However, the proposed method can apply to hybrid systems by modeling GFL devices in a dynamic network, which will be discussed in Section V.

C. System Model and Sensitivity Function

Combining the network and device models, the closed-loop system can be formulated in a feedback-interconnected form as shown in Fig.2(a). Denote by $L(s) = G(s)H_{\text{net}}$ the open-loop transfer function. Then, the closed-loop system is:

$$y = (I + L)^{-1} \cdot \varepsilon + (I + L)^{-1} G \cdot u \stackrel{\text{def}}{=} S \cdot \varepsilon + T \cdot r \quad (6)$$

where the *sensitivity* function $S(s)$ and its complement $T(s)$ quantify the disturbance rejection capability w.r.t. disturbances ε and the tracking performance w.r.t. reference r , respectively. It is noted that these transfer functions are $2n$ -dimension as we consider an n -bus system.

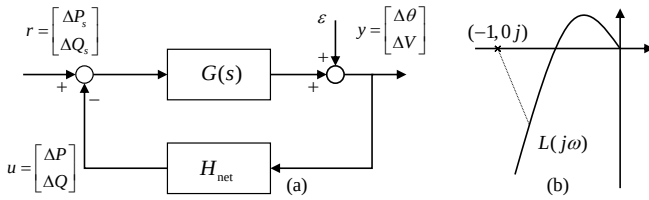


Fig. 2. (a): closed-loop model of power systems, where G and H_{net} denote device and network MIMO dynamics. Both transfer function matrices are $2n \times 2n$ for an n -bus model. (b): geometrical interpretation of the sensitivity function. The absolute value of the sensitivity indicates the inverse of the distance from the generalized Nyquist curve to $(-1, 0j)$.

This paper focuses on the sensitivity function S written as:

$$\begin{bmatrix} \Delta\theta \\ \Delta V \end{bmatrix} = \left(I + \begin{bmatrix} G_{\theta P} H_{P\theta} & G_{\theta P} H_{Q\theta} \\ G_{VQ} H_{Q\theta} & G_{VQ} H_{QV} \end{bmatrix} \right)^{-1} \begin{bmatrix} \Delta\theta_d \\ \Delta V_d \end{bmatrix} \quad (7)$$

The smaller the eigenvalue magnitude of sensitivity functions $\sigma(S(j\omega)) \stackrel{\text{def}}{=} |\lambda(S(j\omega))|$, the better the system's ability to suppress disturbance ε at the frequency ω . Specifically, $\sigma(S(j\omega)) < 1$, or equivalently $\log \sigma(S(j\omega)) < 0$, is desired, indicating that disturbances are not amplified. Alternatively, the sensitivity can be understood from robustness as the distance between the generalized Nyquist curve of $\lambda(L(s))$ and $(-1, 0j)$ satisfies $\text{dist}(-1, \lambda(L(j\omega))) = |1 + \lambda(L(j\omega))| = \sigma(S(j\omega))^{-1}$ as shown in Fig.2(b).

Similar to $\mathcal{G}$, we denote by $\mathcal{G}_n$ the system with the coupling term $H_{\text{net}}^{(c)}$ canceled with corresponding $L_n = GH_{\text{net}}^{(n)}$ and $S_n = (I + L_n)^{-1}$. Physically, the non-coupled system $\mathcal{G}_n$ indicates the *ideal* system where the coupling between phase-angle and voltage dynamics is small enough and ignorable.

Remark 1 (Difference between P - Q and θ - V coupling issues). The power coupling (P - Q coupling) rises from

non-diagonal terms in the transfer mapping from power references $[\Delta P_s, \Delta Q_s]$ to injected powers $[\Delta P, \Delta Q]$ with the relevant closed-loop operator $(I + H_{\text{net}}G)^{-1}H_{\text{net}}G$. See Fig.2(a) for details. By contrast, θ - V coupling refers to non-diagonal terms in the mapping from phase-angle/voltage disturbances $[\Delta\theta_d, \Delta V_d]$ to actual terminal voltage and angle $[\Delta\theta, \Delta V]$ with the relevant closed-loop operator $(I + GH_{\text{net}})^{-1}$. Due to different input/output selections, their transfer matrices couple in different ways and therefore give rise to distinct dynamics and disturbance rejection capability. Hence, the two issues are closely related but not identical.

For power systems in the above formulations, the sensitivity function indicates the internal stability of the feedback system under reasonable conditions, summarized as the following theorem. We again note that this paper focuses on the small-signal stability and is distinct from the large-signal stability analysis based on the Lyapunov techniques [27].

Theorem 1. Let $G(s)$ be a rational, proper, and stable transfer function matrix. Besides, no right-half-plane pole-zero cancellations exist for $G(s)$ and H_{net} . Then, $S(s)$ is stable if and only if the feedback system is internally stable¹, i.e., the following transfer-function matrix is stable:

$$\begin{bmatrix} y \\ u \end{bmatrix} = \begin{bmatrix} (I + GH_{\text{net}})^{-1} & (I + GH_{\text{net}})^{-1}G \\ (I + H_{\text{net}}G)^{-1}H_{\text{net}} & (I + H_{\text{net}}G)^{-1}H_{\text{net}}G \end{bmatrix} \begin{bmatrix} \varepsilon \\ r \end{bmatrix} \quad (8)$$

Proof. It can be found in Appendix A. $\square$

The pole condition of $G(s)$ is easy to satisfy. For properly designed devices, their controllers $G(s)$ should be open-loop stable (note the potential pure integral s in denominators of $\tilde{G}_{\theta P}$ corresponds to the power flow rotation symmetry and thus does not influence the practical stability). Moreover, H_{net} is a constant real matrix. Therefore, this theorem suggests that focusing on $S(s)$ is usually sufficient for the analysis of internal stability, which can be done using well-established theoretical tools such as generalized Nyquist plot, absolute stability, etc. These methods are not the focus of this paper and are thus omitted here.

Before ending this part, we note that the following analysis methods developed based on the sensitivity function are fully model-free and thus applicable to gray-box or even black-box devices. This is because the sensitivity function can be obtained through a standard frequency-scanning technique, which is a common advantage of the frequency-domain methods for the angle-voltage coupling analysis.

D. Classifying Eigenvalues of Sensitivity Function

There are $2n$ eigenvalues for the sensitivity function $S(s)$ of $\mathcal{G}$. We denote the index set of all eigenvalues by $\mathcal{I}$, which can be classified into the phase-angle induced one $\mathcal{I}_A$ and the voltage induced one $\mathcal{I}_V$. It follows that $\mathcal{I} = \mathcal{I}_A \cup \mathcal{I}_V = \{1, 2, \dots, 2n\}$ and $\mathcal{I}_A \cap \mathcal{I}_V = \emptyset$. This paper adopts a homotopic-map-based method introduced in our previous

¹The definition of internal stability here is slightly different from the classical one [30]. However, the equivalence of both is easy to verify.

work. The idea is first proposed in [11] and then developed in [31]. Here, we briefly introduce the idea.

We focus on the network matrix H_{net} as the coupling attributed to its off-diagonal part. Define the homotopic map:

$$\mathcal{H}_{\text{net}}(\mu) = (1 - \mu)H_{\text{net}}^{(n)} + \mu H_{\text{net}} = H_{\text{net}}^{(n)} + \mu H_{\text{net}}^{(c)} \quad (9)$$

It is clear $\mathcal{H}_{\text{net}}(0) = H_{\text{net}}^{(n)}$ and $\mathcal{H}_{\text{net}}(1) = H_{\text{net}}$. Correspondingly, the sensitivity $\mathcal{S}(\mu) = [I + G\mathcal{H}_{\text{net}}(\mu)]^{-1}$ is defined. The matrix $H_{\text{net}}^{(n)}$ and the corresponding S_n are naturally decoupled for phase-angle and voltage dynamics and exhibit a block diagonal structure as shown in (10). As a result, its eigenvalues are naturally classified into angle-related and voltage-related groups.

$$S_n = (I + L_n)^{-1} = \begin{bmatrix} (I + G_{\theta P} H_{P\theta})^{-1} & \\ & (I + G_{VQ} H_{QV})^{-1} \end{bmatrix} \quad (10)$$

After the eigenvalues of $\mathcal{S}(0) = S_n$ are classified, the eigenvalues of $\mathcal{S}(1) = S$ can be classified by tracking their changes when μ increases from 0 to 1. Details regarding the eigenvalue tracking method can be found in [31].

III. SENSITIVITY CONSERVATION AND ITS BREAKING REGARDING ANGLE-VOLTAGE COUPLING DYNAMICS

A. Sensitivity Conservation of Angle-Voltage Coupled Systems

This section aims to identify a quantity that remains conservative in the context of angle-voltage coupling dynamics. To this end, we present the following theorem, which establishes a theoretical foundation for achieving this objective.

Theorem 2. Consider a system as shown in Fig. 2(a) satisfying Assumption 1. Suppose that all entries of its open-loop transfer function $L(s) = G(s)H_{\text{net}}$ are rational, proper, stable and have more than one pole-zero excess. Besides, all singular values of $S(s)$ defined in (6) have constant multiplicity over $s \in \bar{C}_+$, where $\bar{C}_+$ is the closure of the open-right-half plane. If the closed-loop system is stable, then:

$$\sum_{i \in \mathcal{I}} \int_0^\infty \log \sigma_i(S(j\omega)) d\omega = 0 \quad (11)$$

where $\{\sigma_i(s)\}_{i=1}^{2n}$ denotes the (absolute) eigenvalue spectrum.

Proof. The proof can be found in Appendix A. $\square$

Applicability of Theorem 2: The conditions of Theorem 2 are commonly satisfied in practice, as demonstrated below:

- *Open-loop stability.* Noticing the network coupling H_{net} is constant, the characterization of the open-loop transfer function mainly relies on the device dynamics $G(s)$. Any properly designed device should not have open-right-half poles to ensure internal stability. Otherwise, device- or station-wise instability issues might appear, which are beyond the concerns of system-wise stability. Thus, the pole condition generally holds for practical systems and is widely employed in existing literature [32].
- *Relative degree.* The relative degree condition is also easily satisfied, considering the widespread existence of lag effects in inverter-interfaced devices. The inner loop dynamics, including the voltage loop, current loop, and

LCL filter dynamics, always generate control lags, which are usually modeled as an inertia transfer function with one relative degree, see [29] and Appendix B for detailed explanations. Moreover, the measurement, communication, computation, and PWM impulse generation also produce pure time delay T_d , which can be approximated by lags [33]. Combining it with the positive relative degree of control strategies required by the causality, the total relative degree exceeds two and thus meets the requirements. As an example, we note that systems integrated with any controls shown in Table 1 can satisfy the above conditions and thus follow Theorem 2 after properly modeling the inherent dynamic delays.

- *Device compatibility.* Even with significant cross-loop device dynamics ($G_{VP}, G_{\theta Q} \neq 0$), the sensitivity conservation identity still holds once satisfying the theorem conditions. The cross-loop terms change the specific entries of $L(s) = G(s)H_{\text{net}}$ but do not alter the structure of Theorem 2's proof.
- *Network compatibility.* Within the small-signal regime, the sensitivity conservation and the breaking mechanism are inherent. They do not depend on a particular power-flow operating point or on specific load or compensation levels. In other words, heavy or light loading or compensation changes the numerical values of $G(s)$ and $H_{\text{net}}(s)$, but the integral relations and the tradeoffs between phase-angle and voltage sensitivities remain valid, provided Theorem 2 conditions hold.

It is important to clarify that the universality claimed in this paper refers to the proposed method's applicability to any device dynamics that satisfy the conditions of Theorem 2, namely: rational, proper, and stable frequency-domain transfer functions with sufficient relative degree. The method does not presume a specific structure or control strategy for the devices. In other words, it holds regardless of whether the device implements VSG, droop, or other advanced controls. Once the device dynamics meet these conditions, the mechanism remains valid. Meanwhile, as explained above, these conditions are mild and widely satisfied. This ensures that the framework can be directly used for analysis and control design across diverse, heterogeneous, or evolving power systems.

Since Theorem 2 does not assume a specific form of L and the network H_{net} but only relies on some mild requirements, any transfer functions meeting the conditions of the theorem follow this conservation. This feature provides significant applicability to inverter-dominant power systems. Therefore, it also holds for the non-coupled system $\mathcal{G}_n$. In this case, the system's angle and voltage-related subsystems are independent, which yields:

$$\underbrace{\sum_{i \in \mathcal{I}_A} \int_0^\infty \log \sigma_i(S_n(j\omega)) d\omega}_{\mathcal{A}_{\mathcal{I}_A}(S_n)} = \underbrace{\sum_{i \in \mathcal{I}_V} \int_0^\infty \log \sigma_i(S_n(j\omega)) d\omega}_{\mathcal{A}_{\mathcal{I}_V}(S_n)} = 0 \quad (12)$$

Due to the anti-diagonal entries of H_{net} introduced by $H_{\text{net}}^{(c)}$, phase-angle and voltage dynamics are coupled for S . Then, both integrals are usually non-zero. However, since the conditions of Theorem 2 still hold for the coupled models,

as soon as we notice the dynamic models solely rely on the devices, the conservation law holds for the whole coupled system:

$$\underbrace{\sum_{i \in \mathcal{I}_A} \int_0^\infty \log \sigma_i(S(j\omega)) d\omega}_{\mathcal{A}_{\mathcal{I}_A}(S)} + \underbrace{\sum_{i \in \mathcal{I}_V} \int_0^\infty \log \sigma_i(S(j\omega)) d\omega}_{\mathcal{A}_{\mathcal{I}_V}(S)} = 0 \quad (13)$$

For brevity, the four terms in (12) and (13) are respectively denoted as $\mathcal{A}_{\mathcal{I}_A}(S_n)$, $\mathcal{A}_{\mathcal{I}_V}(S_n)$, $\mathcal{A}_{\mathcal{I}_A}(S)$ and $\mathcal{A}_{\mathcal{I}_V}(S)$, where $\mathcal{A}$ denotes the *directed area* following the geometric interpretation of the integral. As indicated by (12) and (13), the coupling between phase-angle and voltage dynamics does not change the total amplitude of the sensitivity integral, indicating the total disturbance rejection capability against phase-angle and voltage dynamics is conservative for general power systems. Then, the sensitivity conservation can be formally characterized as follows.

Sensitivity conservation: For a power system $\mathcal{G}$ satisfying the conditions of Theorem 2, the sensitivity integral of the overall system remains zero whether the coupling is omitted or considered, i.e., $\mathcal{A}_{\mathcal{I}_A}(S_n) + \mathcal{A}_{\mathcal{I}_V}(S_n) = \mathcal{A}_{\mathcal{I}_A}(S) + \mathcal{A}_{\mathcal{I}_V}(S) = 0$.

Remark 2. As a corollary of Theorem 2, there exists an alternative formulation of the sensitivity conservation as follows. A system $\mathcal{G}$ meets the conditions of Theorem 2 satisfies:

$$\sum \Re(p(G)) = \sum \Re(p(S)) \quad (14)$$

where $p(\cdot)$ represents poles of a transfer function. The proof directly follows from combining Theorem 2 and [34]. As the real part of a pole is usually associated with the damping effect, the corollary indicates the coupling does not change the summation of all dampings, but redistributes it through network feedback. The result provides a frequency-domain explanation of [11].

B. Breaking of Sensitivity Conservation in Angle-Voltage Coupling Dynamics

Despite the conservative sensitivity integral across all frequencies, it is redistributed between phase-angle and voltage dynamics, which serves as the mechanism of angle-voltage coupling to influence stability. In fact, We find $\mathcal{A}_{\mathcal{I}_A}(S) = -\mathcal{A}_{\mathcal{I}_V}(S) \neq 0$ always holds in practice. The breaking mechanism is illustrated in Fig.3. We demonstrate the idea step by step as shown below:

- *Single-device non-coupled system*: We start with the simplest situation, i.e., the single-device system without considering the angle-voltage coupling. In this situation, $\mathcal{A}_{\mathcal{I}_A}(S_n) = \mathcal{A}_{\mathcal{I}_V}(S_n) = 0$ holds, indicating that suppressing either the phase-angle or voltage disturbances in one frequency band must be compensated by decreasing the robustness in certain other frequencies. This phenomenon is referred to as “waterbed effect” in the control of single-input-single-output systems [24], [26].
- *Single-device coupled system*: By introducing the coupling term into the non-coupled system, the compensation in the single-device system is extended to not

only between frequencies but also quantities. For example, if the coupling forces the sensitivity redistributed as $\mathcal{A}_{\mathcal{I}_A}(S) = -\mathcal{A}_{\mathcal{I}_V}(S) > 0$, the positive integral value might reduce the disturbance rejection capability of phase-angle dynamics while improving that of voltage dynamics.

- *Multiple-device coupled system*: The analysis can be further extended from single-device systems to multiple-device systems. In this circumstance, the conservation equation (13) implies that sensitivity redistribution can also occur among devices in addition to frequencies and quantities, which is insightful for coordinated control design.

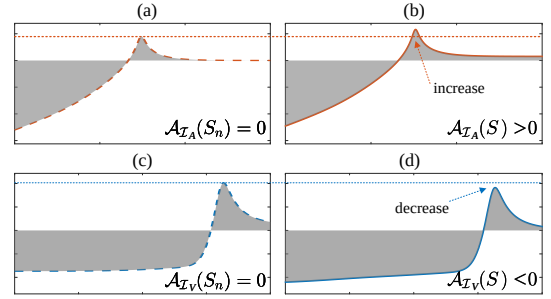


Fig. 3. Illustration of sensitivity breaking. Here, it is an example of $\mathcal{A}_{\mathcal{I}_A}(S_n) = \mathcal{A}_{\mathcal{I}_V}(S_n) = 0$ and $\mathcal{A}_{\mathcal{I}_A}(S) = -\mathcal{A}_{\mathcal{I}_V}(S) > 0$.

The above discussions focus on the integral $\int \log \sigma(S) d\omega$. It generally serves as a metric for the maximum sensitivity $\max \log \sigma(S)$ attributed to the concept of available bandwidth [24]. To show this, we present the following theorem.

Theorem 3. Suppose the conditions of Theorem 2 and the following bounds are satisfied for any i -th eigenvalue:

- $\sigma_i(L(j\omega)) \leq e_i/\omega^{k_i+1} \leq c_i < 1/2, \forall \omega \in [\omega_i^c, \infty)$
- $\sigma_i(S(j\omega)) \leq \alpha_i, \forall \omega \in [0, \omega_i^d]$

where, ω_i^c is the available bandwidth; e_i, k_i, c_i, α_i and $\omega_i^d < \omega_i^c$ are positive constants. Then, the following inequalities hold:

$$\begin{aligned} \max_{\substack{i \in \mathcal{I}_A \\ \omega \in [\omega_i^d, \omega_i^c]}} \log \sigma_i(S(j\omega)) &\geq \frac{\mathcal{A}_{\mathcal{I}_A}(S) + \sum_{i \in \mathcal{I}_A} \left[\omega_i^d \log \frac{1}{\alpha_i} - \frac{3c_i \omega_i^c}{2k_i} \right]}{n \max_{i \in \mathcal{I}_A} (\omega_i^c - \omega_i^d)} \\ \max_{\substack{i \in \mathcal{I}_V \\ \omega \in [\omega_i^d, \omega_i^c]}} \log \sigma_i(S(j\omega)) &\geq \frac{\mathcal{A}_{\mathcal{I}_V}(S) + \sum_{i \in \mathcal{I}_V} \left[\omega_i^d \log \frac{1}{\alpha_i} - \frac{3c_i \omega_i^c}{2k_i} \right]}{n \max_{i \in \mathcal{I}_V} (\omega_i^c - \omega_i^d)} \end{aligned} \quad (15)$$

Proof. It can be found in Appendix A. $\square$

Applicability of Theorem 3: The constants in the theorem are first explained with an illustration in Fig.4. Due to various factors such as under-modeling, sensor noise, and plant bandwidth, the open-loop gain on the high-frequency band above the available bandwidth ω_i^c is desired to decrease for the system's stability and robustness. Usually, the magnitude of ω_i^c is close to the well-known bandwidth of the closed-loop system ω_i^b . The constants e_i and k_i quantify its decreasing

ratio. The second condition is the general design goal, as we usually want the closed-loop system to suppress the disturbances under a given (but also limited) frequency band ω_i^d with the rejection capability quantified by α_i . Therefore, the conditions of Theorem 3 are commonly satisfied.

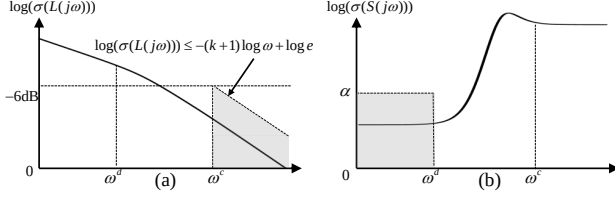


Fig. 4. Illustration of conditions in Theorem 3. (a): the restriction on the open-loop transfer function $L(s)$, i.e., the high-frequency gain should decrease. (b): the restriction on the closed-loop sensitivity $S(s)$, i.e., the low-frequency gain should be bounded.

Theorem 3 suggests that the breaking of sensitivity conservation may worsen the stability of at least one of the quantities. Still take $\mathcal{A}_{\mathcal{I}_A} = -\mathcal{A}_{\mathcal{I}_V} > 0$ as an example. Theorem 3 indicates that $\mathcal{A}_{\mathcal{I}_A} > 0$ increases the lower bound of $\max \log \sigma_i(S(j\omega))$ for $i \in \mathcal{I}_A$. As a result, the weakest mode to reject phase-angle disturbance will be further weakened.

An alternative perspective to understand Theorem 3 is as follows. Noticing the sensitivity conservation $\mathcal{A}_{\mathcal{I}_A} + \mathcal{A}_{\mathcal{I}_V} = 0$, directly sum up two inequalities of Theorem 3 yields:

$$\begin{aligned} & \max_{\substack{i \in \mathcal{I}_A \\ \omega \in [\omega_i^d, \omega_i^c]}} \log \sigma_i(S(j\omega)) + \max_{\substack{i \in \mathcal{I}_V \\ \omega \in [\omega_i^d, \omega_i^c]}} \log \sigma_i(S(j\omega)) \\ & \geq \frac{\sum_{i \in \mathcal{I}} \left[\omega_i^d \log \frac{1}{\alpha_i} - \frac{3c_i \omega_i^c}{2k_i} \right]}{n \max_{i \in \mathcal{I}} (\omega_i^c - \omega_i^d)} \end{aligned} \quad (16)$$

The inequality suggests that the maximum sensitivities of both phase-angle and voltage dynamics have a lower bound determined solely by the system configuration. For example, if decreasing $\max \log \sigma_i(S(j\omega))$ for $i \in \mathcal{I}_V$, the phase-angle-related sensitivity will increase to satisfy the inequality.

Apart from the sensitivity breaking phenomenon, some other interesting results can also be derived from the theorem. For example, if one hopes to achieve better disturbance rejection capability by decreasing α_i in the designable band $[-\omega_i^d, \omega_i^d]$, then the right-hand side of (15) must increase, indicating a larger sensitivity peak beyond the designable region. Meanwhile, an attempt to extend the designable frequency bandwidth ω_i^d closer to the inherent available bandwidth ω_i^c also leads to a larger peak of the sensitivity function. In this context, Theorem 3 lays a *fundamental limitation* on the control design of coupled power systems under given feedback structures [35].

Now, we summarize the breaking of sensitivity conservation induced by angle-voltage coupling dynamics as follows:

Breaking of sensitivity conservation: Despite the invariant total integral of the sensitivity function, the angle-voltage coupling reshapes the sensitivity distribution between two quantities, i.e., $\mathcal{A}_{\mathcal{I}_A}(S) = -\mathcal{A}_{\mathcal{I}_V}(S) \neq 0$. The non-zero integral worsens sensitivity for one quantity while enhancing that for the other, as suggested by (15). In other words, enhancing the stability of one subsystem is at the expense

of the other. More significantly, the restriction cannot be overcome under feedback control structures, which is why it is called *fundamental limitation*.

Therefore, in control designs of angle-voltage coupled systems, it would be tricky to balance the stability of both quantities. The angle-voltage coupling formed by the natural power flow distribution usually worsens the stability, as shown by the following simulations. However, the proposed theory sheds new light on the coordinated control between phase-angle and voltage dynamics. Specifically, as $\mathcal{A}_{\mathcal{I}_A}(S) = -\mathcal{A}_{\mathcal{I}_V}(S)$ holds, we can improve phase-angle/voltage dynamics through proper voltage/phase-angle controller design. This idea is called *cross-loop control* for inverters and has raised increasing concerns [11], [16]. Therefore, while setting the fundamental limitation for the cross-loop controller, the proposed mechanism also provides some principles for its design. A primary example is given in Section IV-B to show how to enhance stability by redistributing angle-voltage coupling.

IV. SIMULATION ANALYSIS

In all simulations of this paper, SGs and constant power loads are replaced by inverter-interfaced resources. The detailed simulation settings are provided in Appendix B. All data and codes are open-sourced at https://github.com/lingo01/Waterbed_Effect to ensure reproducibility.

A. Analysis of Sensitivity Breaking in Single-Device System

To test the proposed theory on the single-device system, we use the well-known VSG control and $Q-V$ droop control, the dynamics of which are provided in Table I. In this case, the sensitivity integral is $\mathcal{A}_{\mathcal{I}_A}(S) = -\mathcal{A}_{\mathcal{I}_V}(S) = 0.2805$. Since phase-angle dynamics dominate the critical eigenvalues, the coupling worsens the overall stability, as shown in Fig. 5(a)(d). According to Theorem 3, $\mathcal{A}_{\mathcal{I}_A}(S) > 0$ will enforce a stability trade-off between phase-angle and voltage subsystems and worsen the former. As confirmed in Fig. 5(b), the coupling amplifies the phase-angle sensitivity peak $\max \log \sigma = 2.0300$ at $10^{0.521} = 3.32$ Hz, which is $10^{2.0171-1.8246} = 1.5575$ times higher than the non-coupled case. Conversely, Fig. 5(c) shows lower voltage-related sensitivity at low frequencies ($\omega < 1$ Hz), confirming the compensation required by sensitivity conservation. The time-domain simulation verifies the result: after counting for the coupling effect, the voltage disturbance is suppressed as shown in Fig. 5(f), while the phase-angle disturbance is slightly amplified as shown in Fig. 5(e).

B. Sensitivity-Redistribution-Based Control Tuning

Despite the sensitivity conservation, one can achieve better dynamic responses by reshaping the sensitivity distribution to a more beneficial direction. Still take the single-device system in Section IV-A as an example, where the coupling has been demonstrated to worsen the phase-angle dynamics. Therefore, the control target is to suppress the phase-angle-induced sensitivity peak around $10^{1 \sim 2}$ Hz. Inspired by the proposed mechanism, we should decrease $\mathcal{A}_{\mathcal{I}_A}(S)$, or equivalently, increase $\mathcal{A}_{\mathcal{I}_V}(S) = -\mathcal{A}_{\mathcal{I}_A}(S)$ to enhance phase-angle

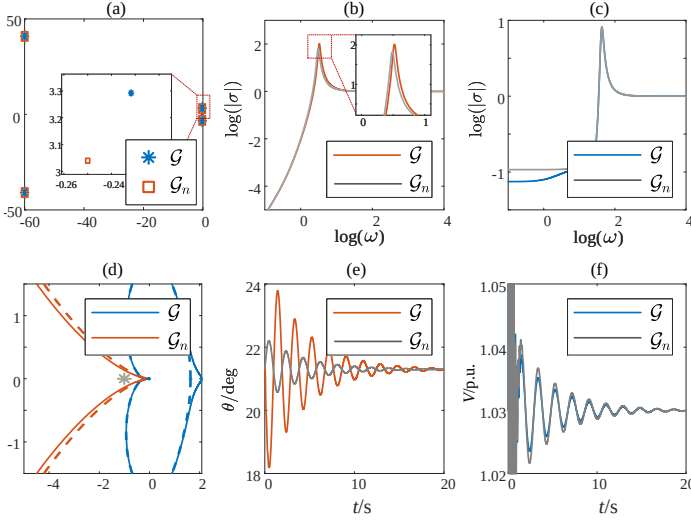


Fig. 5. Breaking of sensitivity conservation in the single-device system. (a): state-space eigenvalues of the linearized closed-loop systems. (b) and (c): eigenvalues of the coupled system S and the non-coupled system S_n w.r.t. the frequencies ω . The eigenvalues in (b) and (c) are induced by phase-angle and voltage dynamics, respectively. (d): generalized Nyquist curves of the coupled system L and the non-coupled system L_n . (e) and (f): time-domain transients of phase-angle and voltage dynamics under disturbance $\varepsilon = 0.1 \cdot [\cos(80^\circ), \sin(80^\circ)] \cdot \text{Impulse}(t)$, where $\text{Impulse}(t)$ is the unit impulse signal.

dynamics since (7) provides cross-loop paths between phase-angle and voltage dynamics. Physically, it suggests that tuning either the phase-angle or voltage control parameters could enhance stability. Both approaches are verified as follows:

- *Phase-angle parameter tuning*: A straightforward way to suppress the maximum sensitivity is to tune the phase-angle dynamics, which can be reformulated as:

$$G_{\theta P} = \frac{1/d}{s(s/(d/M) + 1)} \quad (17)$$

where the electromechanical transients $G_{\text{lag}} \approx 1$. Specifically, we focus on tuning the phase-angle damping coefficient d . As indicated by the generalized Nyquist plot shown in Fig.5(d), increasing d has two aspects of effects. First, the smaller static gain $1/d$ pushes the eigenvalue locus away from $(-1, 0j)$. Second, the bandwidth to hold $\log \sigma_{\mathcal{I}_A}(S(j\omega)) < 0$ as it is controlled by the frequency d/M . Both effects help to decrease the $\mathcal{A}_{\mathcal{I}_V}(S)$ and thus are beneficial to stability. For verifications, the phase-angle damping is doubled, and the test results are shown in Fig.6(a) and (b). After the tuning, the sensitivity area $\mathcal{A}_{\mathcal{I}_A}(S) = -\mathcal{A}_{\mathcal{I}_V}(S)$ decreases from 0.2805 to 0.2736, and the sensitivity peak $\max \sigma(S)$ decreases from 7.5140 to 3.7777. The state-space characteristics also align with the above analysis, where critical eigenvalues move away from the imaginary axis in Fig.6(c).

- *Voltage parameter tuning*: The conservation $\mathcal{A}_{\mathcal{I}_V}(S) = -\mathcal{A}_{\mathcal{I}_A}(S)$ provides an alternative approach to enhance stability through the design of the voltage dynamics $G_{VQ} = 1/k_q$. By decreasing the voltage droop strength k_q , the voltage-related eigenvalues are closer to $(-1, 0j)$ on the Nyquist plot in Fig.5(d). This leads to larger

bandwidth for $\log \sigma_{\mathcal{I}_V}(S(j\omega)) > 0$, and finally, increase $\mathcal{A}_{\mathcal{I}_V}(S)$. For verifications, we consider $k'_q = k_q/2$. Since the corresponding sensitivity is shifted upward, the integral $\mathcal{A}_{\mathcal{I}_V}(S)$ is increased. Fig.6(a) and (b) show the sensitivity distribution before and after tuning. The phase-angle-related sensitivity integral $\mathcal{A}_{\mathcal{I}_A}(S)$ decreases from 0.2805 to 0.2492, while the maximum peak $\max \sigma(S)$ decreases from 7.5140 to 7.0358, verifying the effectiveness of tuning voltage parameters. On the contrary, voltage dynamics are slightly sacrificed on the low-frequency bandwidth due to sensitivity conservation. However, this would be acceptable as the voltage dynamics do not dominate the most critical mode. The state-space eigenvalues depicted in Fig.6(d) verify its effectiveness.

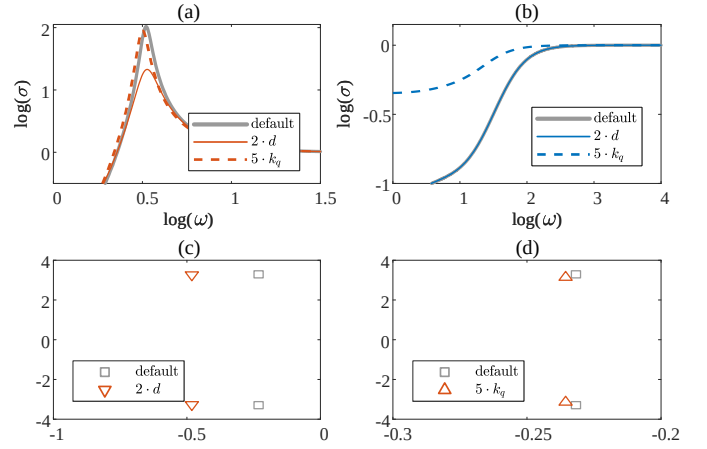


Fig. 6. Redistributing sensitivity through parameter tuning in the single-device system. (a) and (b) are the frequency distributions of phase-angle and voltage-related sensitivity eigenvalues $\sigma(S)$. The gray curves, colored solid curves, and colored dashed curves represent the default, phase-angle-tuned (damping $2 \cdot d$), and voltage-tuned (droop strength $5 \cdot k_q$) tuned parameters in order. (c) and (d) show the state-space eigenvalues before and after phase-angle and voltage tuning, respectively.

Remark 3. A prevalent belief in power system society holds that enhancing a device's phase-angle support capability through approaches like GFM control can improve overall system stability. Furthermore, it is generally assumed that strengthening synchronization performance either has no significant impact on voltage dynamics or at least introduces no adverse effects. However, our study might challenge this faith. In Fig.7, we enhance the synchronization performance by increasing the damping parameter d in the phase-angle control. When d becomes sufficiently large, we have $\Delta\omega_i/\Delta P_i = sG_{\theta P,i}(s) \approx 1/d$, i.e., the device operates approximately in a constant frequency state. In this scenario, Fig.7(a) confirms that the maximum phase-angle-related eigenvalue indeed decreases with increasing d . Nevertheless, while the maximum voltage-related eigenvalue only slightly increased from 1.118653, 1.118726, 1.118940 to 1.118939, their low-frequency disturbance rejection capability exhibits marked degradation as shown in Fig.7(b).

This finding carries critical implications for stability-oriented control design: when attempting to strengthen the stability of one quantity, its effects on other quantities should

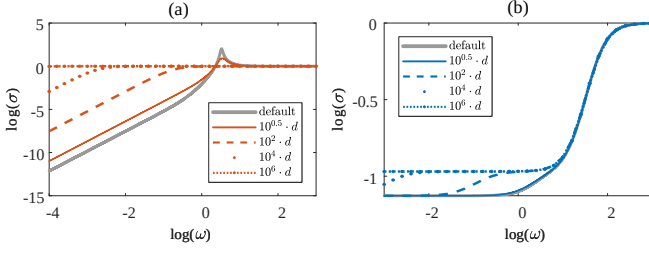


Fig. 7. Sensitivity distribution with ultra-strong phase-angle support. The phase-angle damping is scaled with 10^0 (default), $10^{0.5}$, 10^2 , 10^4 , 10^6 times in order. (a): phase-angle-related eigenvalue of the sensitivity function. (b): voltage-related eigenvalue of the sensitivity function.

be carefully considered. Otherwise, the system might trend toward instability in some unexpected dimensions.

C. Sensitivity Breaking in Multiple-Device System

The theory is further verified on the multiple-device systems such as IEEE 14-, 57-, and 118-bus systems. The sensitivity integral $\mathcal{A}_{\mathcal{I}_A}(S) = -\mathcal{A}_{\mathcal{I}_V}(S) > 0$ in all tested cases indicates the worsened phase-angle dynamics according to the proposed theory. It suggests that the negative impact on phase-angle dynamics might be a common consequence of sensitivity redistribution under coupling. As verified in Fig.8(b), (e), (h) and Table II, the phase-angle-related sensitivity curves are steeper than the non-coupled ones, and their maximum values increase with after considering the coupling. Besides, since the critical modes are more strongly related to the phase-angle dynamics, the increased sensitivity integral worsens the stability of closed-loop systems. On the contrary, the voltage-related sensitivity is suppressed, as shown by the lower blue solid curves compared to the dashed ones in Fig.8(c), (f), and (i). The state-space eigenvalue analysis confirms this fact, as depicted in Fig.8(a), (d), and (g). Compared to the eigenvalues of the non-coupled system (red squares), the consideration of the angle-voltage coupling move even closer to the imaginary axis (blue stars), which aligns with the observations in [11].

TABLE II
MAXIMUM SENSITIVITY VALUE FOR GFM-INTEGRATED SYSTEMS

case	$\mathcal{A}_{\mathcal{I}_A}(-\mathcal{A}_{\mathcal{I}_V})$	$\max \sigma(S)$	$\max \sigma(S_n)$	$\frac{\max \sigma(S)}{\max \sigma(S_n)}$
IEEE 14	3.6739	9.9034	12.7336	1.2235
IEEE 57	21.5448	51.3677	11.1137	4.6010
IEEE 118	15.6626	88.0765	61.5389	1.4315

Time-domain simulation results also support the above analysis, as shown in Fig.9. Disturbances from different directions $[\Delta\theta^T, \Delta V^T]^T = 0.05 \cdot [\cos(\varphi)\mathbb{1}^T, \sin(\varphi)\mathbb{1}^T]^T$ are added to investigate the impact of angle-voltage coupling on the dynamics. The larger φ indicates the disturbance is more voltage-related, while a smaller one suggests the angle-related disturbance. Compared to the non-coupled counterparts in Fig.9(a)-(c), the phase-angle dynamics, after considering the coupling effect, have more oscillation modes with larger amplitude and steady-state error, as shown in Fig.9(d)-(f). Besides, the coupling's amplification effect to the phase-angle oscillation is more significant as the disturbance is more

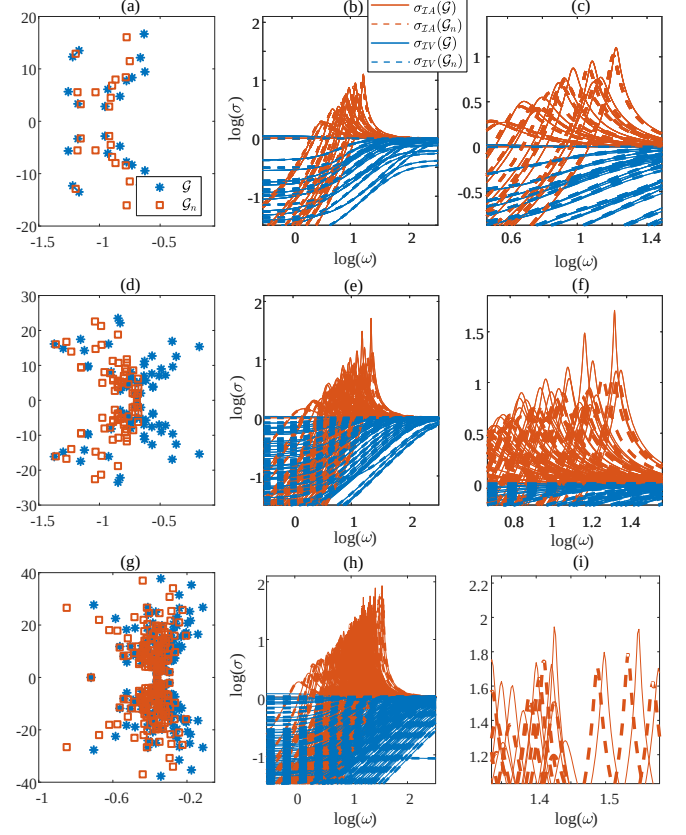


Fig. 8. Breaking of sensitivity conservation in the IEEE 14-, 57, and 118-bus systems. (a), (d) and (g): state-space eigenvalues of the linearized closed-loop models for each benchmark. (b), (e) and (h): eigenvalues of sensitivity functions for these benchmarks. Solid and dashed lines stand for the coupled system S and the non-coupled system S_n , respectively. The orange and blue colors represent the eigenvalues induced by phase-angle and voltage dynamics, respectively. In the figure, we use $\sigma_{\mathcal{I}_A}$ to denote all eigenvalues σ_i for $i \in \mathcal{I}_A$, and similarly for $\sigma_{\mathcal{I}_V}$. (c), (f) and (i) are the zoomed-in versions of (b), (e), and (h).

coupled with voltage dynamics, i.e., $\varphi = 45^\circ$ or 80° compared to $\varphi = 10^\circ$. The simulations conducted on the other test systems provide similar results, but they are not displayed due to space limitations.

D. Verification on Varying Operation Conditions

As analyzed before, the sensitivity conservation and the breaking mechanism are inherent, provided Theorem 2 conditions hold. For validations, we performed a statistical experiment on the IEEE 14-bus benchmark systems. 1,000 random operating points were generated by scaling each active and reactive load. The results are shown in Fig.10. The phase-angle-related and voltage-related integrals preserve the sensitivity conservation across varying operating points. The distribution of $\mathcal{A}_{\mathcal{I}_A} + \mathcal{A}_{\mathcal{I}_V}$ tightly concentrated near 0 with the relative error $1 + \mathcal{A}_{\mathcal{I}_V}/\mathcal{A}_{\mathcal{I}_A} \approx -0.25\%$ induced by the numerical truncation of sampled frequencies. This result indicates that the sensitivity conservation holds robustly across widely varying operation conditions. Besides, the distribution of $\mathcal{A}_{\mathcal{I}_A} = -\mathcal{A}_{\mathcal{I}_V}$ is relatively concentrated w.r.t. operation conditions, suggesting that the integral might be mainly structural.

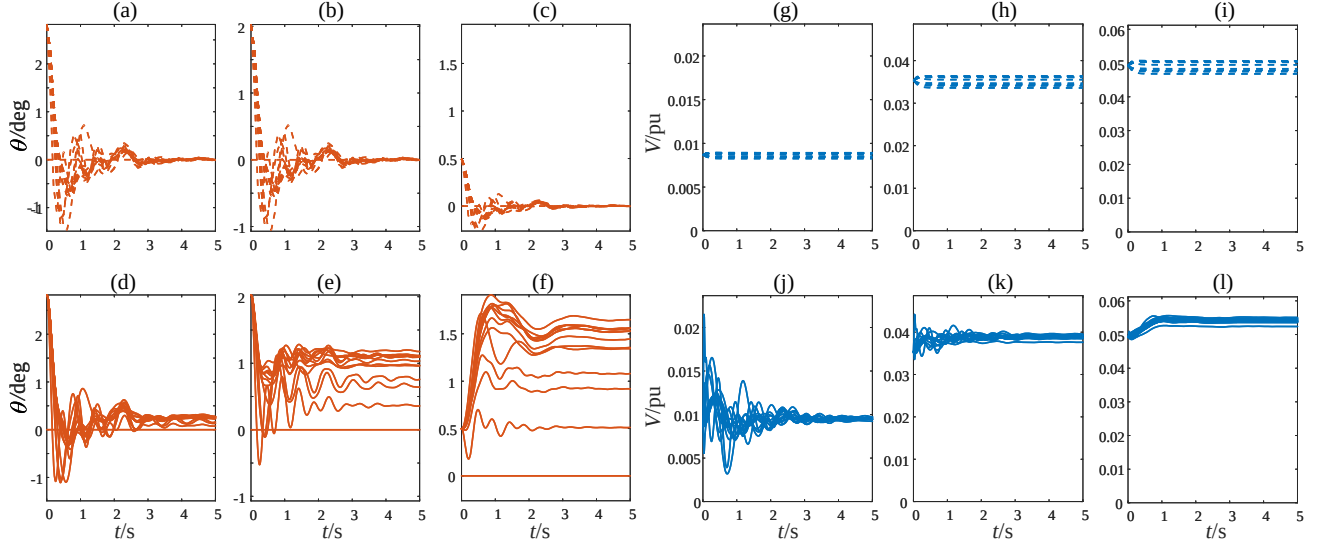


Fig. 9. Time-domain transients of the IEEE 14-bus system under the phase-angle and voltage disturbances. Each i -th device is disturbed with $\Delta\theta_i = 0.05 \cdot \cos(\varphi) \cdot \text{Step}(t)$ and $\Delta V_i = 0.05 \cdot \sin(\varphi) \cdot \text{Step}(t)$ except the reference bus, where φ stand for the disturbance direction and $\text{Step}(t)$ is the unit step signal. (a), (b) and (c): phase-angle dynamics of the non-coupled system $\mathcal{G}_n$ under $\varphi = 10^\circ$, 45° , and 80° . (d), (e) and (f): phase-angle dynamics of the coupled system $\mathcal{G}$ under the same disturbance. (g), (h) and (i) show the non-coupled voltage dynamics, while (j), (k), and (l) depict those of the coupled system, similar to (a)-(f).

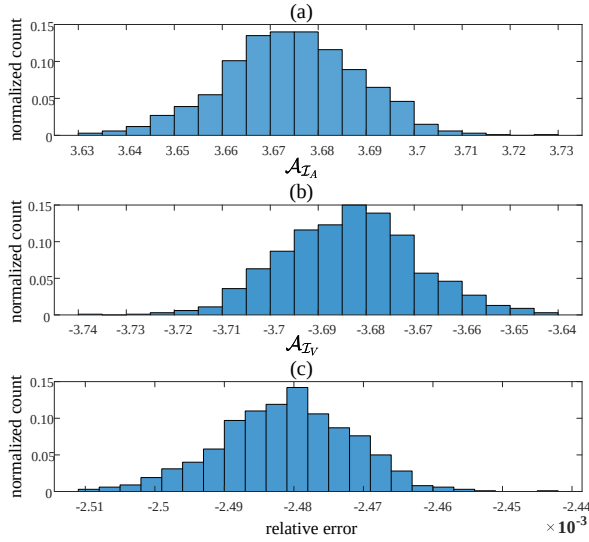


Fig. 10. Statistical results $\mathcal{A}_{\mathcal{T}_A}$ and $\mathcal{A}_{\mathcal{T}_V}$ of the IEEE 14-bus system. A total of 1000 operation conditions are randomly tested, where all active and reactive loads are scaled with relative bias following the normal distribution $\mathcal{N}(0, 0.3^2)$, i.e., the standard deviation $\sigma = 30\%$ compared to the default settings. (a): the phase-angle-related integral $\mathcal{A}_{\mathcal{T}_A}$. (b): the voltage-related integral $\mathcal{A}_{\mathcal{T}_V}$. (c): the relative error $(\mathcal{A}_{\mathcal{T}_A} + \mathcal{A}_{\mathcal{T}_V})/\mathcal{A}_{\mathcal{T}_A} = 1 + \mathcal{A}_{\mathcal{T}_V}/\mathcal{A}_{\mathcal{T}_A}$.

V. EXTENSION TO GFM-GFL HYBRID SYSTEMS

In this section, we extend the proposed method from pure GFM-integrated systems to GFM-GFL hybrid systems.

A. Theoretical Derivation

The sets of GFM and GFL devices are respectively denoted by $\mathcal{M}$ and $\mathcal{L}$, i.e., $\mathcal{V} = \mathcal{M} \cup \mathcal{L}$. Due to the distinguished synchronization mechanism, GFL devices are usually modeled with a different input-output relation from $[\theta, V]$ to $[P, Q]$,

opposite to GFM devices. For $i \in \mathcal{L}$, a classical GFL model with phase-lock-loop (PLL) [36], [37] can be written as:

$$\begin{aligned} -\frac{\Delta P_i}{\Delta\theta_i} &= \frac{d_{p,i}}{\tau_{p,i}s+1} \frac{s(K_{p,i}V_i^*s + K_{i,i}V^*)}{s^2 + K_{p,i}V_i^*s + K_{i,i}V^*} \stackrel{\text{def}}{=} G_{P\theta,i} \\ -\frac{\Delta Q_i}{\Delta V_i} &= \frac{d_{q,i}}{\tau_{q,i}s+1} \stackrel{\text{def}}{=} G_{QV,i} \end{aligned} \quad (18)$$

where K_p and K_i are PI parameters of PLL, and d_p and d_q are $\omega - P$ and $V - Q$ droop gains, respectively. For brevity, we denote $G_{P\theta}(s) = \text{diag}(G_{P\theta,1}, G_{P\theta,2}, \dots, G_{P\theta,n})$ and similar for $G_{QV}(s)$. Then, the dynamics of GFL devices can be represented by $G_{\mathcal{L}} = \text{diag}(G_{P\theta}, G_{QV})$.

The GFL devices cannot be directly written in the form of (5) as the inverse input-output relation leads to non-causality of transfer functions, i.e., the higher numerator's order of $G_{P\theta}^{-1}$ and G_{QV}^{-1} . Alternatively, we turn to treat the GFL devices as a part of the network as they share the same input-output relation, as shown in Fig. 11.

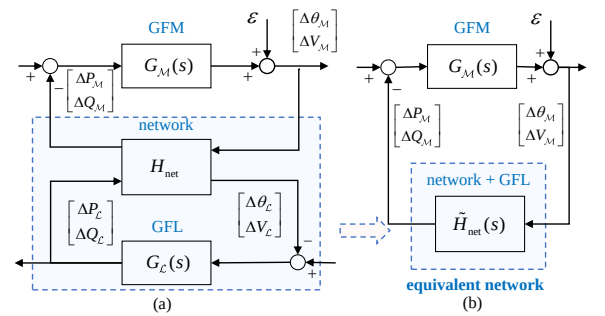


Fig. 11. Network equivalence for hybrid GFM-GFL integrated power systems. (a): model of hybrid GFM-GFL systems. (b): formulation of the equivalent dynamic network by merging the original network and GFL devices.

Then the expression of the equivalent network $\tilde{H}_{\text{net}}$ is derived by the following procedures. The original network

coupling H_{net} for hybrid systems can be block-wise written as (19), where the submatrices denote the corresponding positions in the network coupling matrix. For example, $H_{P\theta, \mathcal{ML}} = [H_{P\theta, ij}]$ where $i \in \mathcal{M}$ and $j \in \mathcal{L}$. Then, following a matrix transformation method, we can obtain its equivalent form from inputs $[\Delta\theta_{\mathcal{M}}^T, \Delta P_{\mathcal{L}}^T, \Delta V_{\mathcal{M}}^T, \Delta Q_{\mathcal{L}}^T]^T$ to outputs $[\Delta P_{\mathcal{M}}^T, \Delta\theta_{\mathcal{L}}^T, \Delta Q_{\mathcal{M}}^T, \Delta V_{\mathcal{L}}^T]^T$.

Meanwhile, the dynamics of GFM devices satisfy $-\Delta P_{\mathcal{L}} = G_{P\theta}\Delta\theta_{\mathcal{L}}$ and $-\Delta Q_{\mathcal{L}} = G_{QV}\Delta V_{\mathcal{L}}$. Combining both above equations and eliminating GFL-related variables using the standard Gaussian elimination method yields:

$$[\Delta P_{\mathcal{M}}^T \quad \Delta Q_{\mathcal{M}}^T]^T = \tilde{H}_{\text{net}}(s) [\Delta\theta_{\mathcal{M}}^T \quad \Delta V_{\mathcal{M}}^T]^T \quad (20)$$

where the equivalent network coupling is provided by:

$$\tilde{H}_{\text{net}}(s) = H_{\mathcal{MM}} - H_{\mathcal{ML}}[G_{\mathcal{L}} + H_{\mathcal{LL}}(s)]^{-1}H_{\mathcal{LM}} \quad (21)$$

The submatrices are defined by the blocks in (19). For example, $H_{\mathcal{ML}}$ is used to denote the following matrix.

$$H_{\mathcal{ML}} = \begin{bmatrix} H_{P\theta, \mathcal{ML}} & H_{PV, \mathcal{ML}} \\ H_{Q\theta, \mathcal{ML}} & H_{QV, \mathcal{ML}} \end{bmatrix} \quad (22)$$

The non-coupled network coupling matrix $\tilde{H}_{\text{net}}^{(n)}$ can be obtained similarly.

Distinguished from the constant matrix H_{net} , the equivalent network $\tilde{H}_{\text{net}}(s)$ is a transfer function as the GFL dynamics $G_{\mathcal{L}}(s)$ are involved. The conditions of Theorem 2 and 3 still hold for the equivalent system, as $\tilde{H}_{\text{net}}$ generally do not provide relative degree (see the inverse structure in (21)). Therefore, through the substitution from H_{net} to $\tilde{H}_{\text{net}}$, the analysis methodology developed for pure GFM-integrated systems is still effective on GFM-GFL hybrid systems by combining (5) and (20).

After obtaining the $\tilde{H}_{\text{net}}(s)$ and $\tilde{H}_{\text{net}}^{(n)}(s)$, the sensitivity transfer matrices can be similarly derived. Here, we note that the replaced sensitivity still reflects the internal stability of the feedback system. In fact, if adding the condition “ $\tilde{H}_{\text{net}}(s)$ is stable” in the proof of Theorem 1, all results derived in this paper still hold.

B. Simulation Verification

Still employ the aforementioned multiple-device benchmarks for verification with detailed simulation settings provided in Appendix B. Their sensitivity integrals are depicted in Table III. For the IEEE 14-, 57-, and 118-bus systems, $\mathcal{A}_{\mathcal{I}_A} = -\mathcal{A}_{\mathcal{I}_V} > 0$ indicates that the coupling worsens the phase-angle dynamics, aligning with the proposed theory. Fig.12 shows their sensitivity distribution. Similar to Fig.8, the phase-angle-related sensitivities (solid curves) are enlarged compared to the non-coupled systems (dashed curves). Besides, the maximum sensitivity increases as listed in Table III since it originates from the phase-angle dynamics. On the contrary, the voltage-related sensitivity is suppressed, as shown by the lower blue solid curves compared to the blue dashed ones.

TABLE III
SENSITIVITY DISTRIBUTION OF GFM-GFL-INTEGRATED SYSTEMS

case name	$\mathcal{A}_{\mathcal{I}_A}$ ($-\mathcal{A}_{\mathcal{I}_V}$)	$\max \sigma(S)$	$\max \sigma(S_n)$	$\frac{\max \sigma(S)}{\max \sigma(S_n)}$
IEEE 14	0.8214	4.8892	4.1752	1.1710
IEEE 57	2.5103	5.8348	5.1632	1.1301
IEEE 118	27.6237	7.6169	6.1517	1.2382

VI. CONCLUSION

This paper provides a novel frequency-domain perspective to understand the mechanism of angle-voltage coupling's impact on system dynamics. While the sum of sensitivity integrals is conservative, the coupling redistributes them between phase-angle and voltage dynamics, thus worsening one quantity's dynamics while improving the other's. Therefore, the stability of the coupled system is usually weakened, akin to the waterbed effect. The sensitivity conservation and breaking laws cannot be directly bypassed by any controllers with a generic feedback structure, as shown in Fig.2(a). In other words, this phenomenon should be treated as a fundamental limit of coupled power systems, as recognized for the first time in the angle-voltage coupling dynamics.

Despite the conservative total sensitivity, the proposed theory indicates that we might still improve the weakest dynamics by judiciously transferring sensitivity between frequencies, devices, or between phase-angle and voltage dynamics. Besides, this design should be comprehensively evaluated, as strengthening the synchronization capability might also lead to detrimental effects on voltage dynamics, as revealed by our studies. Therefore, the sensitivity conservation and its breaking laws should be carefully treated as one design principle for system controls.

Future works are twofold: First, generalize the proposed methods to non-minimal-phase systems using Poisson-like integral results [25], which further enables consideration of non-minimal-phase power system dynamics such as network line dynamics in electromagnetic timescales. Second, apart from tuning the existing controllers as presented above, the proper cross-loop controls [11] might also benefit the redistribution of sensitivity, the systematic design of which is our future research direction.

APPENDIX A PROOF OF THEOREMS

Proof of Theorem 1. The stability of the left-upper matrix of (8) equals to that of $(I + GH_{\text{net}})^{-1} = S(s)$. Besides, as the stability of $G(s)$ and no pole-zero cancellation have been assumed in the conditions, $(I + GH_{\text{net}})^{-1}G$ is stable if and only if $S(s)$ is stable. Hence, we bridge the stability of the right-upper matrix of (8) with that of $S(s)$. Finally, consider the matrix identity $(I + H_{\text{net}}G)^{-1}H_{\text{net}} = H_{\text{net}}(I + GH_{\text{net}})^{-1} = H_{\text{net}}S$. It directly links the two lower matrices of (8) with $S(s)$. Similarly, as H_{net} is constant, they are stable if and only if $S(s)$ is stable. Therefore, the above analysis yields that the stability of the transfer matrix (8) is equivalent to that of the sensitivity $S(s)$. $\square$

$$\begin{bmatrix} \Delta P_M \\ \Delta P_L \\ \Delta Q_M \\ \Delta Q_L \end{bmatrix} = \begin{bmatrix} H_{P\theta,MM} & H_{P\theta,ML} & H_{PV,MM} & H_{PV,ML} \\ H_{P\theta,LM} & H_{P\theta,LL} & H_{PV,LM} & H_{PV,LL} \\ H_{Q\theta,MM} & H_{Q\theta,ML} & H_{QV,MM} & H_{QV,ML} \\ H_{Q\theta,LM} & H_{Q\theta,LL} & H_{QV,LM} & H_{QV,LL} \end{bmatrix} \begin{bmatrix} \Delta\theta_M \\ \Delta\theta_L \\ \Delta V_M \\ \Delta V_L \end{bmatrix} \quad (19)$$

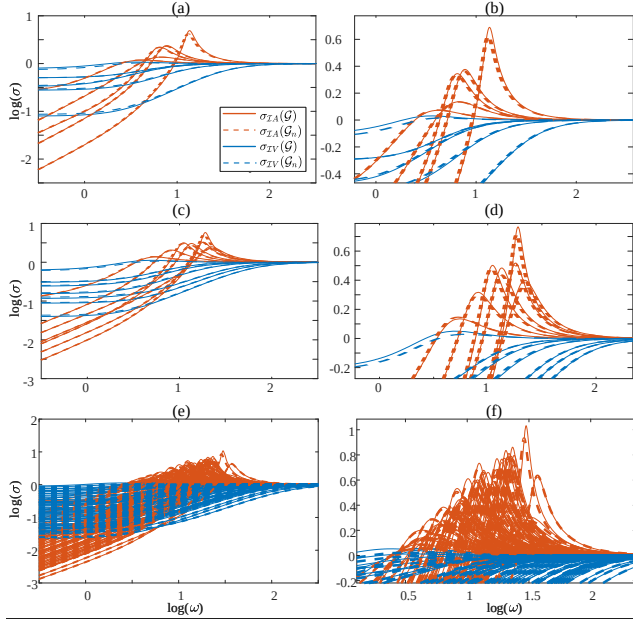


Fig. 12. Breaking of sensitivity conservation in the IEEE 14-, 57-, and 118-bus GFM-GFL hybrid systems. (a), (c) and (e): sensitivity eigenvalues. Solid and dashed lines represent the coupled system S and the non-coupled system S_n , respectively. (b), (d) and (f) are the zoomed-in versions of (a), (c), and (e).

Proof of Theorem 2. The result can be viewed as a corollary of the main theorem in [25], a more systematic statement of which can be found in [26, Theorem 4.2.2], which says that under the given conditions, each singular value of the sensitivity matrix satisfies:

$$\int_0^\infty \log \delta_i(S(j\omega)) d\omega = \frac{1}{2} \iint_{\bar{C}^+} x \nabla^2 \log \delta_i(S(x+jy)) dx dy \quad (23)$$

where δ denotes the singular value and $\bar{C}^+$ is the closure of the open-right-half plane. Then, sum all singular values, and we obtain:

$$\sum_{i=1}^{2n} \int_0^\infty \log \delta_i(S(j\omega)) d\omega = \frac{1}{2} \iint_{\bar{C}^+} x \nabla^2 \log \det(S(x+jy)) dx dy \quad (24)$$

As the conditions of the theorem are satisfied, $\det(S)$ is analytic on $\bar{C}^+$, otherwise, the closed-loop transfer function is unstable. Then $\log \det(S)$ is also analytic on $\bar{C}^+$. Hence, we obtain zero for the equation according to the Cauchy-Goursat theorem. Finally, the proof will be completed by noticing $\sum \log \delta_i(S(j\omega)) = \log \det(S(j\omega)) = \sum \log \sigma_i(S(j\omega))$ holds for any $\omega \geq 0$. $\square$

Proof of Theorem 3. Due to the symmetry of phase-angle and voltage dynamics, we denote $\mathcal{I}_*$ for either $\mathcal{I}_A$ or $\mathcal{I}_V$ in this

proof. The integral of any i -th sensitivity eigenvalue ($i \in \mathcal{I}_*$) can be split into three terms written as:

$$\int_0^\infty \log \sigma_i(S(j\omega)) d\omega = \left(\int_0^{\omega_i^d} + \int_{\omega_i^d}^{\omega_i^c} + \int_{\omega_i^c}^\infty \right) \log \sigma_i(S(j\omega)) d\omega \quad (25)$$

The first term can be bounded by $\textcircled{1} \leq \omega_i^d \log \alpha_i$. The second and third terms respectively satisfy:

$$\int_{\omega_i^d}^{\omega_i^c} \log \sigma_i(S(j\omega)) d\omega \leq \max_{r \in \mathcal{I}_*} (\omega_r^c - \omega_r^d) \max_{i \in \mathcal{I}_*} \log \sigma_r(S(j\omega)) \quad (26)$$

$$\int_{\omega_i^c}^\infty \log \sigma_i(S(j\omega)) d\omega \leq \int_{\omega_i^c}^\infty |\log(1 + \sigma_i(L(j\omega)))| d\omega \quad (27)$$

To examine the effect of the third term, we use the following fact, proof of which can be found in [26, Example A.8.6].

$$|\log(1 + s)| \leq 3|s|/2, \quad |s| < 1/2 \quad (28)$$

As $\sigma_i(L(j\omega)) \leq e_i/\omega^{k_i+1} \leq c_i < 1/2$ for $\forall \omega \in [\omega_i^c, \infty)$ has been assumed in the condition, the above fact implies:

$$\begin{aligned} \int_{\omega_i^c}^\infty \log \sigma_i(S(j\omega)) d\omega &\leq \frac{3}{2} \int_{\omega_i^c}^\infty \log \sigma_i(L(j\omega)) d\omega \\ &\leq \frac{3}{2} \int_{\omega_i^c}^\infty \frac{e_i}{\omega^{k_i+1}} d\omega = \frac{3e_i}{2k_i(\omega_i^c)^{k_i}} \leq \frac{3c_i\omega_i^c}{2k_i} \end{aligned} \quad (29)$$

Then, adding together all eigenvalues within the indicator set $\mathcal{I}_*$ and using the above bounds will complete the proof. $\square$

Before ending the section, we note that the bound provided in Theorem 3 can be squeezed if additional information regarding the system is given. For example, for the phase-angle dynamics with VSG control, we already know the phase-angle related eigenvalue $\log(\sigma_{\mathcal{I}_A}) \rightarrow -\infty$ as $\log(\omega) \rightarrow -\infty$. Therefore, the restriction on the designable frequency band can be more precisely described, says $\log(\sigma_i(S(j\omega))) \leq \beta_i \log(\omega) + \gamma_i, \forall i \in \mathcal{I}_A, \omega \in [0, \omega_i^d]$, where β_i and γ_i are given constants. Based on these restrictions, a better approximation than (15) can be obtained. However, it is noted that the provided theorem takes advantage of a uniform description for any type of device dynamics.

APPENDIX B SIMULATION SETTINGS

All data and codes are open-sourced at https://github.com/lingo01/Waterbed_Effect to ensure reproducibility.

A. Power Grid Settings

The static data of all benchmark systems, including network topology, bus types, branch parameters, and default power flow, are identical to the MATPOWER standard datasets used in the power-systems community [38]. Their single-line diagrams can be found in [39] and some typical ones are shown in Fig. 13.

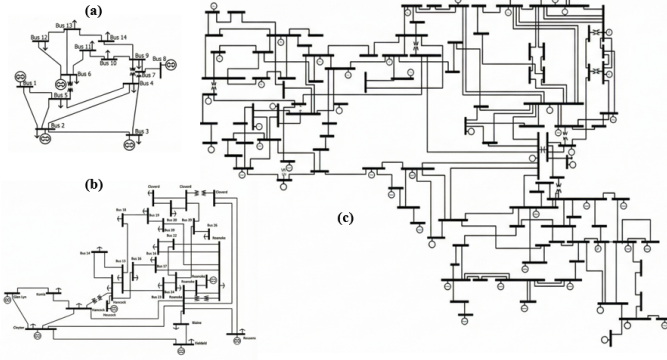


Fig. 13. Diagram of the IEEE benchmark systems. (a): IEEE 14-bus system. (b): IEEE 30-bus system. (c): IEEE 118-bus system. The diagrams are redrawn from [39].

B. GFM-Integrated Single-device System

The equilibrium of the device is set as $(\theta_s, V_s) = (21.3^\circ, 1.03)$ while the infinite bus satisfies $(\theta_0, V_0) = (0^\circ, 1)$. The GFM devices take the strategies of VSG and $Q-V$ droop controls. The dynamics are provided in Table I. The parameters of the network and device are $R + jX = 0.2 + j0.3$ and $(M, d, T_{il}, \tau, k_q^{-1}) = (0.3, 0.15, 0.01, 0.05, 0.8)$.

C. GFM-Integrated Multiple-device Systems

In the simulations, all SGs and constant power loads are replaced by GFM devices. The network topology and power flow data are borrowed from the MATPOWER standard dataset. The GFM devices take the strategies of VSG (or $P-f$ droop with low-pass power measurement filters) and $Q-V$ droop controls. The dynamics are provided in Table I. Parameters $(M, d, T_{il}, \tau, k_q^{-1})$ are selected from normal distribution $\mathcal{N}(0.3, 0.1^2)$, $\mathcal{N}(0.8, 0.1^2)$, $\mathcal{N}(0.02, 0.002^2)$, $\mathcal{N}(0.3, 0.1^2)$ and $\mathcal{N}(0.8, 0.2^2)$, referring to [10], [11].

D. Hybrid GFM-GFL Integrated Multiple-device Systems

All generators are replaced with GFM inverters, and all loads are replaced with GFL inverters. The network topology, power flow data, and GFM devices are set with the same method as above pure GFM-integrated systems, except for strengthened $k_q^{-1} \sim \mathcal{N}(0.2, 0.05^2)$ to guarantee the system stability. As for the GFM devices, their dynamics are provided by (18). The GFL parameters $(K_i, K_p, \tau_p, \tau_q, d_p, d_q)$ are selected from $\mathcal{N}(590, 30^2)$, $\mathcal{N}(34, 5^2)$, $\mathcal{N}(1.0, 0.2^2)$, $\mathcal{N}(1.0, 0.2^2)$, $\mathcal{N}(1.5, 0.3^2)$, $\mathcal{N}(1.5, 0.3^2)$, which refer to [37].

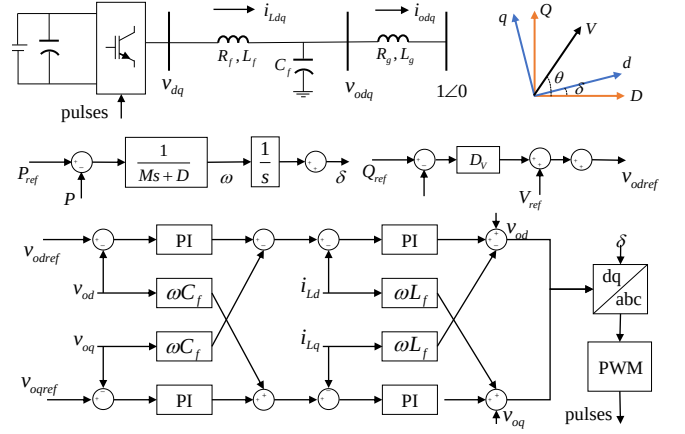


Fig. 14. Electromagnetic control loops of inverters.

E. Electromagnetic Dynamics

In this paper, we consider the electromagnetic control loops of inverters as shown in Fig. 14.

As we focus on the transmission grid stability, the timescale of the current loop is much faster than the voltage loop and the synchronous dynamics of the system [2]. Therefore, the current loop dynamics can be approximated by $G_{cc} \approx 1$. Then, the voltage loops commonly use the PI controllers, i.e.,

$$G_{vc} = K_{vp} + \frac{K_{vi}}{s} \quad (30)$$

where K_{vp} and K_{vi} are the PI parameters of voltage loops. Then, the lag effect of electromagnetic loops can be expressed as [29]:

$$\frac{\Delta V_d(s)}{\Delta V_{dref}(s)} = \frac{\Delta V_q(s)}{\Delta V_{qref}(s)} = \frac{1}{1 + \frac{C_f s}{G_{vc} G_{cc}}} = \frac{K_{vp} s + K_{vi}}{C_f s^2 + K_{vp} s + K_{vi}} \quad (31)$$

where C_f is the LCL filter capacitor. The reference voltage V_{dref} is the output signal of $\tilde{G}_{VQ}$ and V_{dref} is usually set as zero. Therefore, it holds that:

$$-\frac{\Delta V}{\Delta Q} = -\frac{\Delta V_d}{\Delta V_{dref}} \cdot \frac{\Delta V_{dref}}{\Delta Q} = \frac{K_{vp} s + K_{vi}}{C_f s^2 + K_{vp} s + K_{vi}} \cdot \tilde{G}_{VQ} \quad (32)$$

In other words, the lag effect of electromagnetic transients is:

$$G_{lag} = \frac{K_{vp} s + K_{vi}}{C_f s^2 + K_{vp} s + K_{vi}} \quad (33)$$

Alternatively, if considering the time delay induced by the power measurement filter, it can be written as:

$$G_{lag} = \frac{1}{T_{il} s + 1} \quad (34)$$

Throughout the simulation, we adopt $K_{vp} = 0.5$, $K_{vi} = 5$, $C_f = 1.59155 \times 10^{-6}$ and $T_{il} = 0.02$.

F. Disturbance Settings

In the multi-machine time-domain simulations of this paper, we apply step disturbances in phase angle and voltage (i.e., steps in $\Delta\theta_d$ and ΔV_d). The discrepancy from ΔV_d to ΔV , and similarly from $\Delta\theta_d$ to $\Delta\theta$, can model local reference

changes and unmodeled inner dynamics of devices, measurement/actuation lags and inaccuracies, and related effects. They are standard tests in small-signal stability analysis of power systems [16].

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